Affine Grassmannian elements of type C_n^(1) are parametrized by self-conjugate 2n-cores

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/sigma-2025-023-macdonald-identities-weyl-kac-denominator-fo/pinned-original-source.tex (Macdonald Identities, Weyl-Kac Denominator Formulas and Affine Grassmannian Elements); no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The author asserts the stability step ('One first checks') without executing it; the descent crux carries its reason clause ('because c is self-conjugate') and the strictly-decreasing descent terminates, which is implicit. The census printed field 7.1 is the published-version coordinate; the card face renders Lemma 5.3 by source-text counter stepping (census v3.5 rule). No numbered external references are invoked inside the span.

Editorial note. The label Lemma_selfconjugate names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own Lemma environment in the article source.

Editorial note. Journal-unit assembly: card-local preamble (author theorem environments and macros verbatim), article class, no bibliography in the counted span (the span carries no \cite commands).

Editorial note. Revision r002: the r001 card-local preamble was malformed - the lane's preamble extractor dropped five structural lines of the author's \blfootnote definition (\begingroup, \addtocounter, \endgroup, the closing brace, a bare % comment), leaving an unbalanced brace and an xelatex failure; the extractor is fixed and the preamble is regenerated verbatim from the pinned source. Additionally \usepackage{makecell} is declared via assembly.extra_packages because the source preamble redefines the journal's tablehead macros (\theadalign, \theadfont, \theadgape, \cellgape via \Gape) that the article class does not define; no card-body text changed.

Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- The action of the affine Grassmannian elements W_{C_n^(1)} on core partitions via the raised operators s_i^{C_n^(1)} (section 5 setup).

- The set C_{2n}^s of self-conjugate 2n-cores and the notion of removable nodes of a partition.

Lemma 5.3. *We have $W_{C_n^{(1)}} \cdot \varnothing = \mathcal{C}_{2n}^s$, that is, the affine Grassmannian elements of type $C_n^{(1)}$ are parametrized by the self-conjugate $2n$-cores.*

Proof. One first checks that $\mathcal{C}_{2n}^s$ is stable under the action of $W_{C_n^{(1)}}$. Next, it suffices to observe that for any nonempty $c \in \mathcal{C}_{2n}^s$, there exists at least an integer $i = 0, \ldots, n$ such that all i and $2n - i$ nodes of c are removable because c is self-conjugate. This implies that $s_i^{C_n^{(1)}} \cdot c = c'$ where $c' \neq c$ belongs to $\mathcal{C}_{2n}^s$ and has a number of nodes strictly less than c. □